MIL-A-48121 (MU)
1 November 1972
SUPERSEDING
(See 6.5)

# MILITARY SPECIFICATION

## ADAPTER, SAFETY WITH BOOSTER FOR FUZE, PD, M720 LOADING, ASSEMBLING AND PACKING

### 1. SCOPE

1.1 This specification covers the loading, assembling and packing for one type of safety adapter designated as Adapter, Safety with Booster for Fuze, PD, M720.

### 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified.

SPECIFICATIONS

MILITARY

MIL-A-2550 - Ammunition and Special Weapons; General Specification for
MIL-I-45607 - Inspection Equipment, Supply and Maintenace of

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-650 - Explosives, Sampling, Inspection, Testing
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

FSC: 1390

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DRAWINGS

ARMY

9229637 - Safety Adapter Assembly with Booster for Fuze, PD, M720

PUBLICATIONS

ARMY

ET-9229637 - Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

3. REQUIREMENTS

3.1 Material. - Materials shall be in accordance with applicable drawings and specifications.

3.2 Parts and Assemblies. - All parts and assemblies shall comply with all requirements specified on the applicable drawings and with all requirements in the applicable specifications.

3.3 Workmanship. - All parts shall be fabricated and finished in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, chips, dirt, grease, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. All required markings shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 General Quality Assurance Provisions. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.



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4.1.1 Submission of Product. - At the time the completed lot of product is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified within this specification.
b. Number of units of product inspected.
c. Results obtained, by defect code, for all inspections performed.
d. Drawing, specification number and date, together with an identification and date of changes thereto.
e. Certificates of conformance on all material purchased by the contractor when such material is controlled by Government or commercial specifications referenced in any of the contractual documents.
f. Number of items in the lot.
g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 Inspection Provisions

4.2.1 Lot Formation. - The term "lot" as used throughout this specification refers to an inspection lot which is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance of the lot with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions. Changes to the process, specifications or drawings not affecting safety, performance, interchangeability or storage, as determined by the Government, shall not be deemed to alter the homogeneity of the lot. Inspection lots shall comply with MIL-STD-105. Unless otherwise approved by the Contracting Officer, the inspection lot size of end items deliverable under the contract shall be not less than the smallest weekly estimate of quantities contractually scheduled for production during the contract period, nor more than the largest monthly quantity contractually scheduled for delivery during the contract period. Inspection lots of components or subassemblies, other than the items of delivery, shall be homogeneous and of a size



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mutually convenient to both the contractor and the Government inspector. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each lot shall contain:

a. Metal parts of one interfix lot number from one manufacturer.
b. Booster pellets of one interfix lot number from one manufacturer.
c. Tetryl of one interfix lot number from one manufacturer.

4.2.2 Examination. - Inspection for Critical Defects (and Major Defects, when so specified) shall be 100 percent. Sampling plans and procedures for Major and Minor Defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to individual characteristics listed using an AQL of 0.65 percent for each Minor Defect and an AQL of 0.40 percent for each Major Defect, except where 100 percent inspection is specified. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.2.2.1 Safety Adapter Assembly, Prior to Assembly of Booster Pellet and Cup (see dwg. 9229637).

| Categories | Defects | Method of Inspection | Code No. (see 6.2) |
|---|---|---|---|
| Critical: | | | |
| 1. | Solid tetryl bridge from one booster cup thread to another ..... | Visual | 01001 |
| 2. | Rotor not locked in unarmed position ..... | Visual | 01002 |
| Major: | AQL 0.65 Percent | | |
| 101. | Booster closing cup assembly missing ..... | Visual | 01003 |
| 102. | Booster pellet missing ..... | Visual | 01004 |
| 103. | Foreign matter in booster cup (prior to insertion of pellet into cup) ..... | Visual | 01005 |
| Minor: | AQL 0.65 Percent | | |
| 201. | Evidence of poor workmanship (see 3.3) ..... | Visual | 01006 |



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4.2.2.2 Closing Cup Assembly (see dwg. 8595506 covering a detail of dwg. 9229637).

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: | None defined. | | |
| Major: | AQL 0.65 Percent | | |
| 101. | Booster closing cup not having a full 360 degrees (°) crimp ..... | Visual | 02001 |
| 102. | Booster closing cup explosive above flush ..... | Gage | 02002 |
| 103. | Booster closing cup explosive below surface of body ..... | Gage | 02003 |
| Minor: | AQL 0.65 Percent | | |
| 201. | Evidence of poor workmanship (see 3.3) ..... | Visual | 02004 |

4.2.2.3 Safety Adapter Assembly (see dwg. 9229637).

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: | None defined. | | |
| Major: | AQL 1.0 Percent | | |
| 101. | Threads damaged ..... | Visual-Gage | 03001 |
| 102. | Total length, maximum (max.) ..... | Gage | 03002 |
| 103. | Diameter of booster cup, max. ..... | Gage | 03003 |
| 104. | Booster cup loosens when an unscrewing torque of 35 inch-pounds is applied ..... | Gage | 03004 |
| 105. | Foreign matter ..... | Visual | 03005 |
| Minor: | AQL 4.0 Percent | | |
| 201. | Staking missing or inadequate ..... | Visual | 03006 |
| 202. | Booster cup cracked or split ..... | Visual | 03007 |
| 203. | Marking misleading or illegible ... | Visual | 03008 |
| 204. | Booster pellet moves when assembly is shaken ..... | Manual | 03009 |
| 205. | Dust cover missing or loose ..... | Visual | 03010 |
| 206. | Concentricity of booster cup with body diameter above cup ..... | Gage | 03011 |



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| | | | |
|---|---|---|---|
| 207. | Booster cup contacting shoulder of body ..... | Visual | 03012 |
| 208. | Evidence of poor workmanship (see 3.3) ..... | Visual | 03013 |

4.3 Testing

4.3.1 Booster Pellet (see dwg. 8595510)

4.3.1.1 Binder - Lubricant Content of Tetryl - Major Defect - Code No. 06001. - A sample of 20 grams shall be required from each batch to determine the content of binder-lubricant of the tetryl. This test shall be performed as specified in 4.4.1. Failure of the sample to meet the requirements specified on the applicable drawings shall be cause for rejection of the respective batch.

4.3.1.2 Specific Gravity of Booster Pellet - Major Defect - Code No. 06002. - A sample of 10 booster pellets shall be selected from each lot of booster pellets to determine the specific gravity. This test shall be performed as specified in 4.4.2. Failure of one or more booster pellets to comply with the requirements specified on the applicable drawing shall be cause for rejection of the lot.

4.3.2 Closing Cup Assembly (see dwg. 8595506)

4.3.2.1 Graphite Content of Tetryl - Major Defect - Code No. 07001. - A sample of 20 grams shall be taken from each batch of tetryl. The test shall be performed as specified in 4.4.3. Failure of any sample to comply with the requirement specified on the applicable drawing shall be cause for rejection of the respective batch.

4.3.2.2 Specific Gravity of Charge - Major Defect - Code No. 07002. - A sample size of 20 loaded closing (or lead) cups shall be selected from each lot for determination of specific gravity. The test shall be performed as specified in 4.4.4. Failure of any sample to comply with the requirement on the applicable drawing shall be cause for rejection of the lot.

4.3.3 Inspection Equipment. - The inspection equipment required to perform the examinations and tests prescribed in this section is identified by Equipment Tabulation ET-9229637 (see 6.3).

4.3.3.1 Government Rights to Documentation. - Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection,



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overhaul, shipping, storage, identification of stock, ordering and storage of replacement of parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.3.3.2 Supply and Maintenance. - Supply and maintenance of the equipment listed in the ET shall be in accordance with Specification MIL-I-45607.

4.3.3.3 Government Use of Contractor's Inspection Equipment. - The contractor shall make available to the Government all inspection equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment and verification of its accuracy shall be supplied by the contractor for the performance of examination or test by the Government.

4.4 Test Methods and Procedures

4.4.1 Binder - Lubricant Content of Tetryl in Booster Pellet. - An accurately weighed portion of five (5) grams of the sample shall be transferred to a tared medium porosity sintered glass filtering crucible. The sample shall be extracted with four (4) 10 milliliter portions of acetone. Each portion shall be maintained in contact with the residue for approximately five (5) minutes before applying suction. Any lumps which may persist shall be broken up. The sides of the crucible shall be washed down with 10 ml each portion of acetone to insure complete removal of the tetryl. The residue shall be aspirated until the odor of acetone is no longer detectable. The crucible and contents shall be placed in a drying oven at 105° plus or minus 5° centigrade (C) for 1/2 hour, cooled at room temperature in a desiccator and weighed. The above extraction, drying and weighing procedure shall be repeated with individual 10 ml portions of acetone until the loss in weight on subsequent extractions totals less than 0.002 gram. Calculate the total percentage of binder-lubricant as follows:

$$\text{Total Percent of Binder-Lubricant} = \frac{100A}{B}$$

where:

A = Weight of residue in cricible.
B = Weight of specimen on a moisture free basis.

4.4.2 Specific Gravity of Booster Pellet. - Specific gravity of booster pellet shall be determined according to Method 203.1 of MIL-STD-650.

4.4.3 Graphite Content of Tetryl in Closing Cup. - The test for the determination of graphite content in the closing cup charge shall be the same as described in 4.4.1. Calculate the total percentage of graphite content as follows:



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$$\text{Total Percent of Graphite Content} = \frac{100A}{B}$$

where:

A = Weight of resudie in cricible.
B = Weight of specimen on a moisture free basis.

4.4.4 Specific Gravity of Closing (Lead) Cup Charge. - The loaded cup shall be weighed in air and the weight shall be recorded as A. The loaded cup shall be weighed in water at a temperature of 25°C and the weight shall be recorded as B. The volume of the loaded cup shall be determined by subtracting B from A (A-B) and the volume shall be recorded as C. The tetryl composition shall be removed from the cup by washing the cup with acetone. The empty cup shall be weighed in air and the weight shall be recorded as D. The empty cup shall be weighed in water at 25°C and the weight shall be recorded as E. The volume of the empty cup shall be determined by subtracting E from D (D-E) and the volume recorded as F. The volume of the tetryl shall be determined by subtracting the volume of the empty cup (F) from the volume of the loaded cup (C), i.e., (C-F), and the volume shall be recorded as G (C-F = G). The weight of the tetryl composition shall be determined by subtracting the weight of the empty cup in air (D) from the loaded cup in air (A), and the weight shall be recorded as H (A-D = H). Specific gravity of the charge shall be calculated as follows:

$$\text{Specific Gravity} = \frac{H}{G}$$

where:

G = The volume of tetryl composition.
H = The weight of tetryl composition.

## 5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, labeling and marking shall be as specified on dwg. 7548174 and 7548175.

## 6. NOTES

6.1 Ordering Data. - Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Ammunition data cards shall be prepared for each lot in accordance with MIL-STD-1167.



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6.2 Inspection Code Numbers. - The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection Equipment Designs. - Inspection equipment designs are of two types - Government Special Inspection Equipment (SIE) designs and contractor designs. SIE designs are designated by drawing numbers on the Equipment Lists referenced on the Equipment Tabulation. Design responsibility for all other inspection equipment is assigned to the contractor. Equipment to be designed by the contractor is designated "CD". The contractor need not furnish any design when a complete Government SIE design is provided. Unless otherwise specified, however, the contractor may submit alternate or modified contractor designs of SIE in accordance with 6.3.2 and 6.3.3 should he elect to do so.

6.3.1 SIE Designs. - SIE designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.

b. A source control drawing or a specification control drawing as defined in MIL-STD-100.

c. An envelope drawing, as defined in MIL-STD-100, which establishes the criteria which a detail design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.3.2 Contractor Designs. - Contractor designs are required for all inspection equipment for which SIE designs are not specified and may include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public.) Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be perpared for any special fixture(s) required to be used with commercial equipment or with SIE designs if not otherwise covered thereby (see 6.3.1.c). Designs shall be of the category and form (per MIL-D-1000 specified in the Contract Data Requirements Lists (DD Form 1423). The contractor is referred to MIL-HDBK-204, "Inspection Equipment Design" for guidance. The specification number and the applicable five-digit defect code number (or other specific identifying information) from Section 4 of this specification shall be referenced on each contractor design, together with the component or assembly drawing number and revision letter to which the specific design applies.

6.3.3 Submission of Designs for Approval. - Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment.



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Designs shall be submitted for approval in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements List (DD Form 1423) or in the contract. Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will normally be accomplished within one month after receipt.

6.4 Visual Examination.- When compliance with applicable requirements is in doubt, as a result of visual examination this characteristic may be measured to determine acceptability.

6.5 This specification includes the requirements of PA-PD-3089 dated 26 January 1968.

Custodian:
Army-MU

Preparing Activity
Army-MU

Project Number: 1390-A-001



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